The default direction of a rotation is the anti-clockwise direction. The angle of rotation can be any value between 0 and 360 degrees, both are included.

Rotation of 360° is called a full turn, rotation of 180° is called a half turn, rotation of 90° is called a quarter turn.

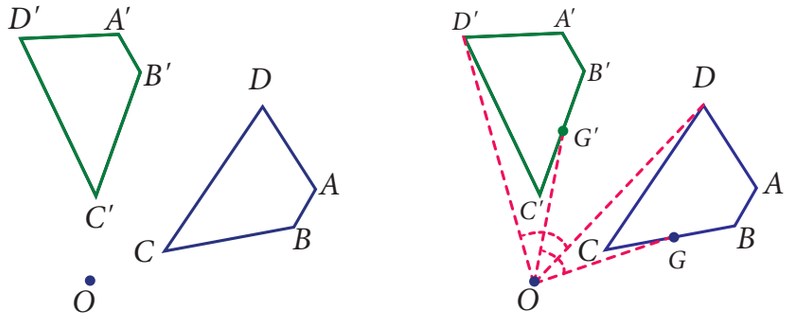
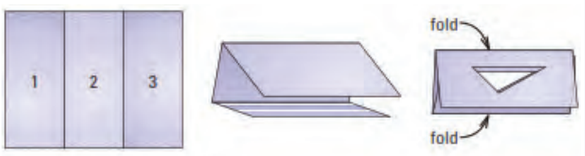


Fig. 4.12

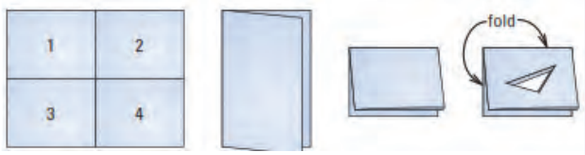
In *Fig. 4.12* the preimage ABCD is rotated about the point O to get the image A'B'C'D'. Here the angles ∠AOA', ∠BOB', ∠COC', ∠DOD' are equal. Any point G on the preimage ABCD will have a corresponding image G' on A'B'C'D' such that ∠GOG' = ∠AOA' = ∠BOB' = ∠COC' = ∠DOD' which is the angle of rotation.

Activity

- Fold a piece of paper and lable it as shown. Cut scalene triangle out of the folded paper and unfold the paper. You can see triangles in all three parts How are the triangles in parts 2 and 3 are related to triangle in part 1?



- Fold a piece of paper and lable it as shown. Cut scalene triangle out of the folded paper and unfold the paper. You can see triangles in all four parts. How are the triangles in part 2, part 3 and part 4 are related to triangle in part 1?



How to rotate a shape about a point?

To rotate a shape about a point with the given angle, we have to rotate every vertex individually and connect them again.

Here ΔABC is rotated about O with angle of 100°.

Step1. Draw CO. Make angle of 100° with vertex C and side CO using a protractor.

Step2. Use a compass to construct CO' = CO

Step3. Locate A' and B' in the similar way.

Step4. Join A', B', C' to form ΔA'B'C'

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